

Q1. Which register in RISC-V always holds the value zero?

- x1
- x31
- x0
- x10

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Q2. What is the base opcode format for R-type instructions in RV32I?

- 0x33
- 0x13
- 0x03
- 0x23

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Q3. Which instruction is used for unconditional jumps in RV32I?

- beq
- jal
- j
- jr

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Q4. How is memory-mapped I/O accessed in RISC-V?

- Through special registers
- Using direct instructions
- Using standard load/store
- Not possible

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Q5. How many general-purpose registers does RV32I have?

- 8
- 16
- 32
- 64

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Q6. Which of the following is NOT an RV32I instruction?

- addi

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- lw
- sll
- mul

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Q7. What is the opcode for the 'addi' instruction in RV32I?

- 0x33
- 0x13
- 0x63
- 0x03

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Q8. What does the 'lui' instruction do?

- Loads an immediate into upper bits
- Loads data from memory
- Links and jumps
- Logical AND

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Q9. What is the width of address bus in a 32-bit RISC-V system?

- 16
- 24
- 32
- 64

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Q10. How many independent channels does AXI protocol have?

- 2
- 3
- 5
- 7

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Q11. Which of the following is a valid AXI channel?

- Read Address
- Write Address
- Write Data

- All of the above

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Q12. Which of the following statements about AXI is true?

- AXI is not pipelined
- AXI supports only read operations
- AXI allows out-of-order transactions
- AXI does not support burst transfers

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Q13. What does AXI stand for in AMBA protocols?

- Advanced eXtensible Interface
- Advanced Exchange Interface
- Addressable External Interface
- Advanced External Interconnect

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Q14. In AXI protocol, what is the purpose of the 'AWVALID' signal?

- To indicate address is valid for reading
- To indicate address is valid for writing
- To validate data
- To acknowledge transfer

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Q15. In FPGA design, what is the purpose of a constraints file (XDC)?

- To configure peripherals
- To define timing and pin mappings
- To initialize memory
- To compile Verilog

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Q16. What is the primary benefit of AXI4 over AXI3?

- Support for only fixed burst
- Reduced bandwidth
- Higher throughput with burst transfers
- More control signals

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Q17. Which Vivado stage is responsible for placing logic blocks onto FPGA fabric?

- Synthesis
- **Implementation**
- Bitstream generation
- IP Packaging

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Q18. Which of the following is a correct sequence in FPGA flow?

- Synthesis -> Simulation -> Implementation -> Bitstream
- Simulation -> Implementation -> Bitstream -> Synthesis
- **Specification -> Synthesis -> Implementation -> Bitstream**
- Bitstream -> Simulation -> Synthesis

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Q19. How does AXI-Lite differ from AXI4?

- **No burst support**
- Supports only read
- No handshaking
- Only for memory-mapped RAM

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Q20. What does the 'BVALID' signal indicate in AXI?

- Address validity
- Burst end
- **Write response is valid**
- Byte enable

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Q21. What is typically used to bridge CPU and memory on FPGA?

- PLL
- DMA
- **AXI interconnect**
- UART

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Q22. In AXI, what is the role of WSTRB signal?

- Indicates write start

- Byte-wise write strobe

- Burst enable
- Stream signal

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Q23. What is a main benefit of using AXI over older bus standards?

- Fewer signals
- Out-of-order support and parallelism
- Only synchronous
- Non-deterministic behavior

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Q24. What FPGA resource is typically used for synchronous block memory?

- LUT
- BRAM
- DSP
- FF

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Q25. Which AXI signal ensures the slave is ready to accept data?

- AWREADY
- WVALID
- ARREADY
- WDATA

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Q26. Which HDL construct helps infer BRAM during synthesis?

- assign
- case
- if-else
- always @(posedge clk) with enable

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Q27. Which Verilog construct models clock edge sensitivity?

- always @(posedge clk)
- initial
- assign

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- module

Q28. Which tool is typically used for simulating Verilog before synthesis?

- Vivado
- Quartus
- ModelSim
- Xilinx SDK

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Q29. What is the main difference between RTL and Gate-level simulation?

- Gate-level ignores timing
- RTL is post-synthesis
- Gate-level includes timing delays
- RTL includes bitstream

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Q30. In FPGA design, what is the typical result of synthesis?

- Bitstream
- Netlist
- Executable
- Source code

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